



Solar Optimism Needs To Be Back In India

Sustainable growth of the solar energy industry depends on successful management of photovoltaic (PV) waste. PV waste recycling is still at a nascent stage globally, both in terms of technical standards and physical infrastructure. Use of potentially-hazardous materials in manufacturing and lack of commercially-viable module recycling technologies demand a strong regulatory approach to this problem



Sudeshna Das
is director at
ComConnect
Consulting

India has the potential to become the world leader in the solar sector by satisfying future energy demands while reducing carbon footprints. Growing need for energy in India is perfectly aligned with its green energy transition initiatives. However, the country needs to become self-reliant by developing the value chain ecosystem. This, in turn, will also help the Indian solar sector to become competitive in the long run. The industry can ensure long-term sustainable growth on successful adoption of emerging business models, innovative technologies like floating solar as well as with the proper vision to handle future challenges like PV waste management.

Market at a glance

The first quarter of 2019 saw sluggish growth due to general elections as well as challenges like Goods & Service Tax (GST) implementation, safeguard duty and compliance with

Bureau of Indian Standards (BIS) standards, sudden change in solar park development policy, under subscription and cancellation of tenders, no implementation of much heralded schemes like Sustainable Rooftop Installation for Solar Transfiguration of India (SRISTI) and Kisan Urja Suraksha Evam Utthaan Mahabhiyan (KUSUM), and so on.

According to a Bridge To India (BTI) report, total installed utility-scale solar installation (Fig. 1) in the country reached 26,154MW as on March 2019. Capacity addition fell to 4810MW, down a steep 47 per cent (Fig. 2) over the previous year.

However, total pipeline of utility-scale solar projects is quite high. Thus, installation capacity is expected to boom in the next 24 months.

Last year was notable for the exponential growth in the open access (OA) market, where total OA installed capacity reached approximately 1359MW by March 2018 (Fig. 3) itself. Karnataka led the market in terms of installed capacity (1592MW), followed by Madhya Pradesh (332MW), Telangana (322MW), Andhra Pradesh (208MW) and Tamil Nadu (164MW).

OA solar is fundamentally a very attractive market opportunity. Main drivers are rising cost advantage over grid power and policy thrust on mitigation in carbon emissions. Solar power also enjoys several operational advantages over thermal and wind power such as shorter gestation period, lower technical complexity and larger potential area of deployment.

Solar park schemes simplified

Modifications in Development of Solar Parks and Ultra Mega Solar Power Project scheme

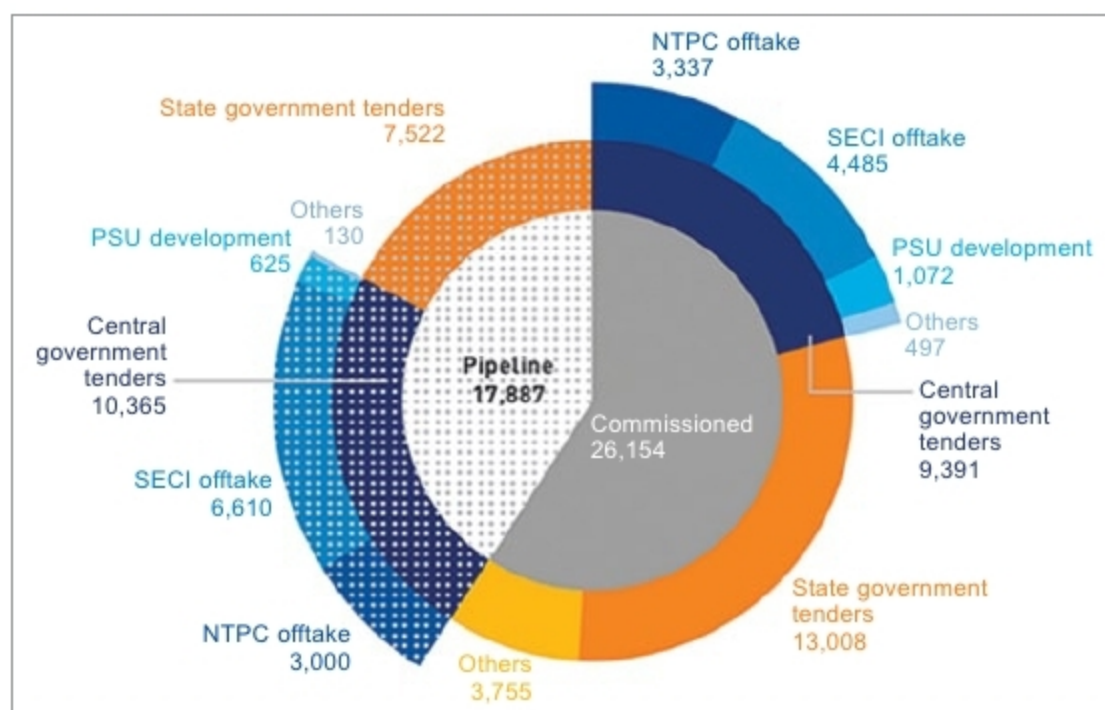


Fig. 1: Total utility-scale solar installations in India in 2018-19 (Source: Bridge To India)